

Final Project Report Template

1. Introduction

1.1. Project overviews This project focuses on analyzing agricultural crop production in India. The main objective is to study crop production, area, yield, different crops, states, and yearly changes using data analysis and visualization.

1.2. Objectives

To study and analyze crop production in different states of India and understand the changes in crop production over the years.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Reference: <https://mno.com/templates/customer-problem-statement/>

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer	Understand crop production patterns in different states of	It is difficult to compare production	Agricultural data contains information from many	Confused and uncertain while making decisions



		India	of different crops and states	states, crops, and years	
PS-2	An agricultural planner	Analyze crop production trends across India	It is difficult to identify important patterns from large amounts of data	The data contains many crops, states, years, and production values	Overwhelmed and less confident in decision making

2.2. Project P proposal (Proposed Solution)

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To study and analyze crop production in different states of India and understand the changes in crop production over the years.
Scope	The project covers crop production data from different states and years. It will show crop-wise, state-wise production using charts and dashboards.
Problem Statement	
Description	India produces many types of crops in different states. It is difficult to understand and compare all the crop production data manually. A simple data analysis and visualization system is needed to understand the data easily.
Impact	This analysis can help users understand which crops are produced more, which states have higher production, and how crop production changes over time.
Proposed Solution	
Approach	First, collect the agricultural crop production data. Then clean and organize the data. After that, analyze the data and create charts and an interactive dashboard to show the results clearly.
Key Features	1.State-wise crop production 2.Crop-wise production 3.Year-wise production

	4.Top-production crops 5.Top-production states 6.production trends 7.Interactivechaers and filters 8.Easy-to-understand dashboard
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computer/Laptop for data analysis	Normal laptop/PC
Memory	RAM required for analysis	8 GB
Storage	Storage for data and project files	256 GB or more
Software		
Frameworks	Tools used for analysis and visualization	Tableau
Libraries	Libraries for data cleaning/analysis	Python, Pandas
Development Environment	Software used for project work	Tableau / Jupyter Notebook
Data		
Data	Agricultural crop production dataset	CSV/ Excel dataset

2.3. Initial Project Planning

Initial Project Planning Template

Date	06 September 2026
Team ID	-
Project Name	India's Agricultural Production Analysis
Maximum Marks	4 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection and extraction	USN-1	Story 1. Downloading dataset	2	Easy	Mahaveer	20 August 2026	28 August 2026
			Story 2. Loading the dataset	1	Easy			
Sprint-1	Data Preparation	USN-2	Story 3. Clean the data	3	Moderate	Mahaveer	20 August 2026	28 August 2026
Sprint-2	Documentation and Recoding	USN-3	Story 7. Creating demo link	3	Moderate	Mahaveer	29 August 2026	06 September 2026
			Story 8. Creating documents	5	Difficult			
Sprint-1	Data	USN-4	Story 4. Developing Visualization	5	Difficult	Mahaveer	20 August	28 August

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Data Collection Plan Template

Section	Description
Project Overview	This project focuses on analyzing agricultural crop production in India. The main objective is to study crop production, area, yield, different crops, states, and yearly changes using data analysis and visualization.
Data Collection Plan	The data be collected from government agricultural data sources such as the directorate of Economics and Statistics, Department of Agriculture and farmers welfare and the open government data platform of India.
Raw Data Sources Identified	The raw data will contain information about crop name, state, district, year, season, cultivated areas, production, and crop yield. These data will be used for cleaning, analysis, and visualization.

Raw Data Sources Template

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	India's Agricultural crop production Analysis	https://docs.google.com/spreadsheets/d/1SLqjCAhYMldwNg__f2sZfVty8vz84mnM/edit?usp=sharing&oid=117180477350188983562&rtf=of=true&sd=true	CSV	442MB	Public
...	...	...	...	...	...

3.2. Data Quality Report

Data Collection and Preprocessing Phase

Date	06 September 2026
Team ID	
Project Title	India's Agricultural crop production Analysis
Maximum Marks	3 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
Dataset	Some records may have missing values in crop production, area, or yield columns.	Moderate	Check missing values using Excel.
...	...	...	...

3.3. Data Exploration and Preprocessing

Date	06 September 2026
Team ID	
Project Title	India' Agricultural crop production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset contains information about crop production in India, including year, district, crop, season, area, production and yield.
Data Cleaning	Check missing values and duplicate records, remove duplicates, handle missing values, and correct formatting errors.
Data Transformation	Filter and sort data by year, state, crop and season. Create useful, calculated columns for analysis.
Data Type Conversion	Convert year, area, production and yield into the correct data types for analysis.
Column Splitting and Merging	Split or merge columns only when required.
Data Modeling	Create measures for production and yield.
Save Processed Data	Save the cleaned dataset as PDF form.

4. Data Visualization

4.1. Framing Business Questions

4.2. Developing Visualizations

Business Question and Visualization Report

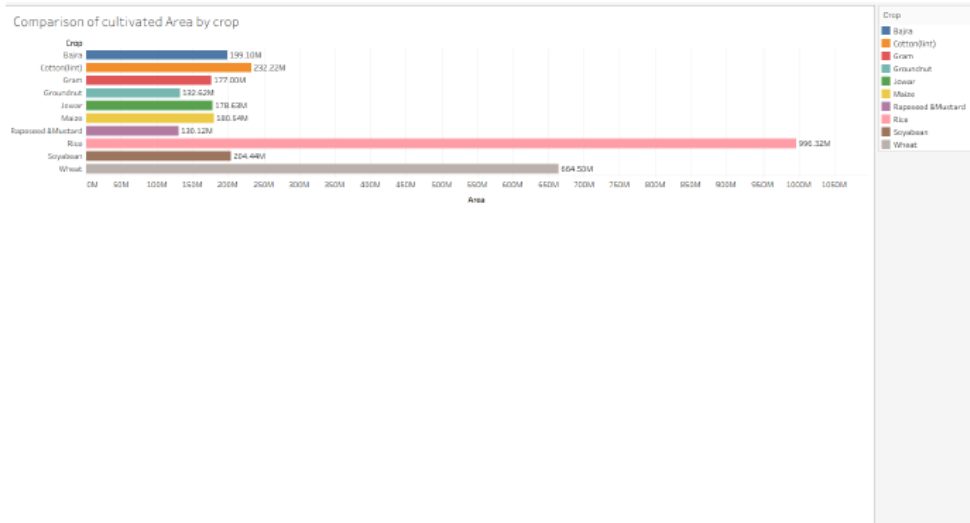
Date	06 September 2026
Team ID	
Project Name	India's Agricultural crop production Analysis
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting.

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

1. Which crops have the highest production in India?
 - o **Visualization:** Bar chart showing production by crop.
 - o **Screenshot of visualization:**



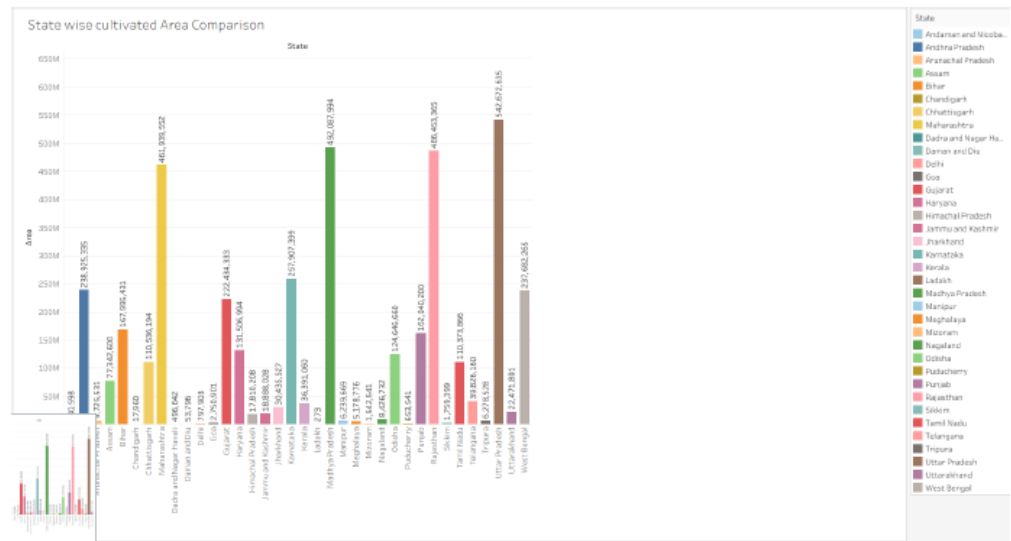
2. How has crop production changed over the years?

- Visualization: Line chart showing yearly production.
- Screenshot of visualization:



- o **Visualization:** Bar chart showing production by state.

- Screenshot of visualization:



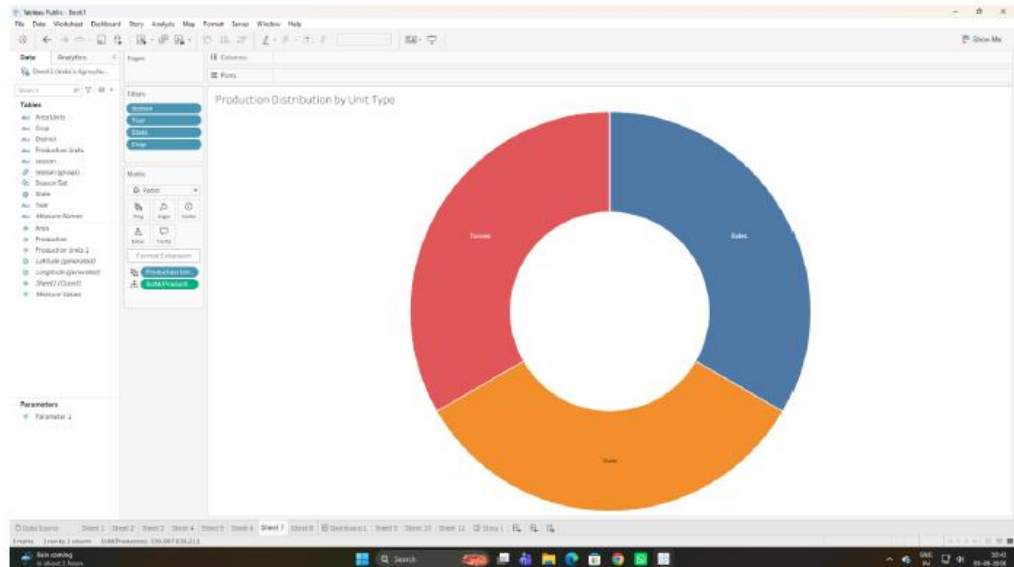
4. Which crops are grown most in different seasons?

- o **Visualization:** pai chart showing production by season.

- Screenshot of visualization:

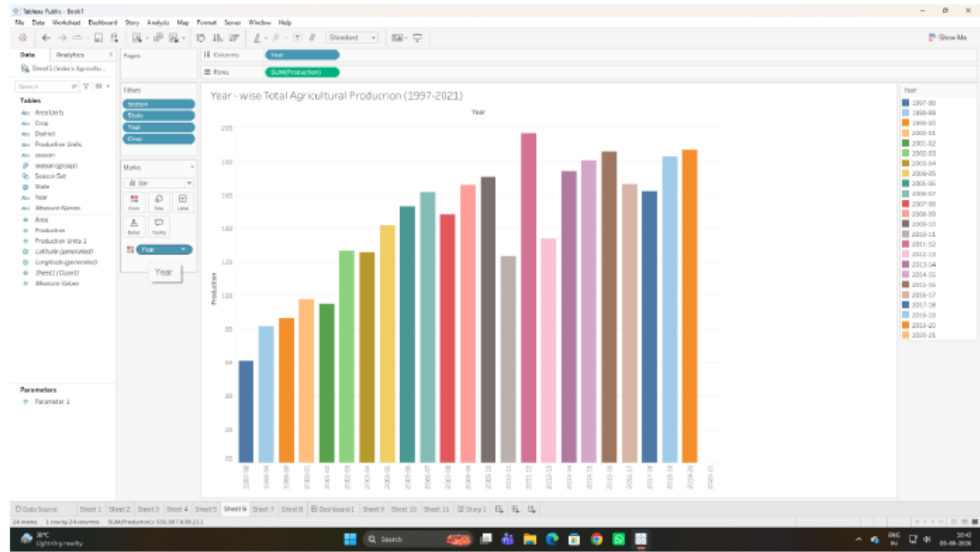
4. Which crops are grown most in different seasons?

- Visualization: pai chart showing production by season.
- Screenshot of visualization:



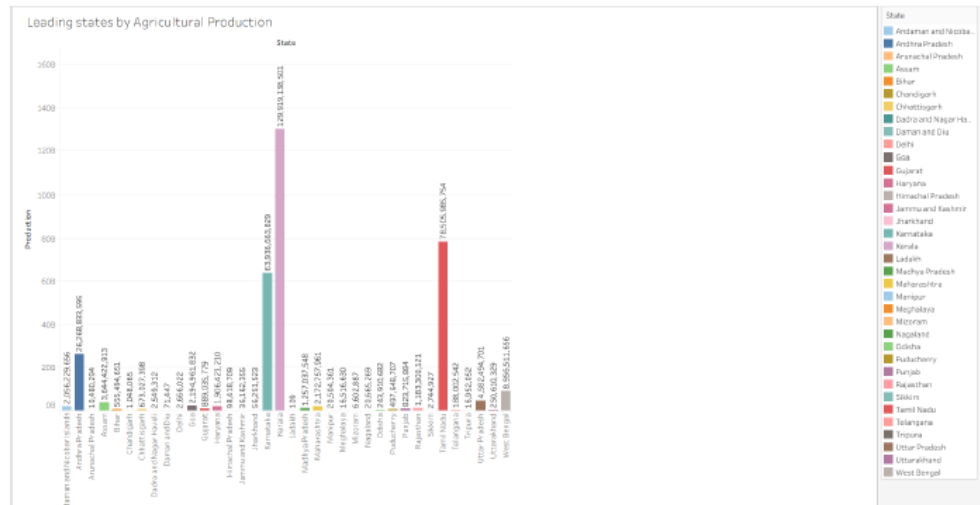
5. Which year has high agricultural production?

- Visualization: Bar chart showing average yield by crop.
- Screenshot of visualization:



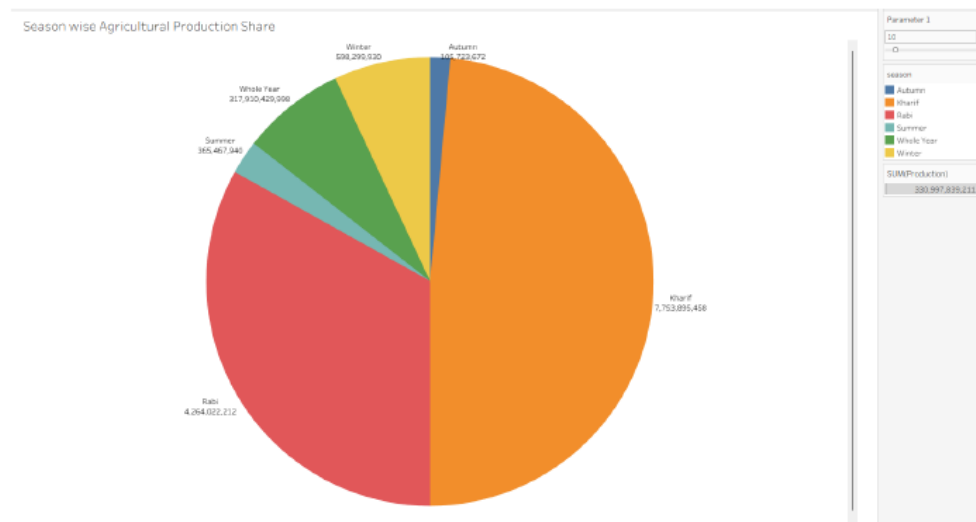
6. Which states have the highest cultivated area?

- o Visualization: Bar chart showing cultivated area by state.
- o Screenshot of visualization:



7. Which season has most agricultural production ?

- **Visualization:** season wise agricultural share
- **Screenshot of visualization:**

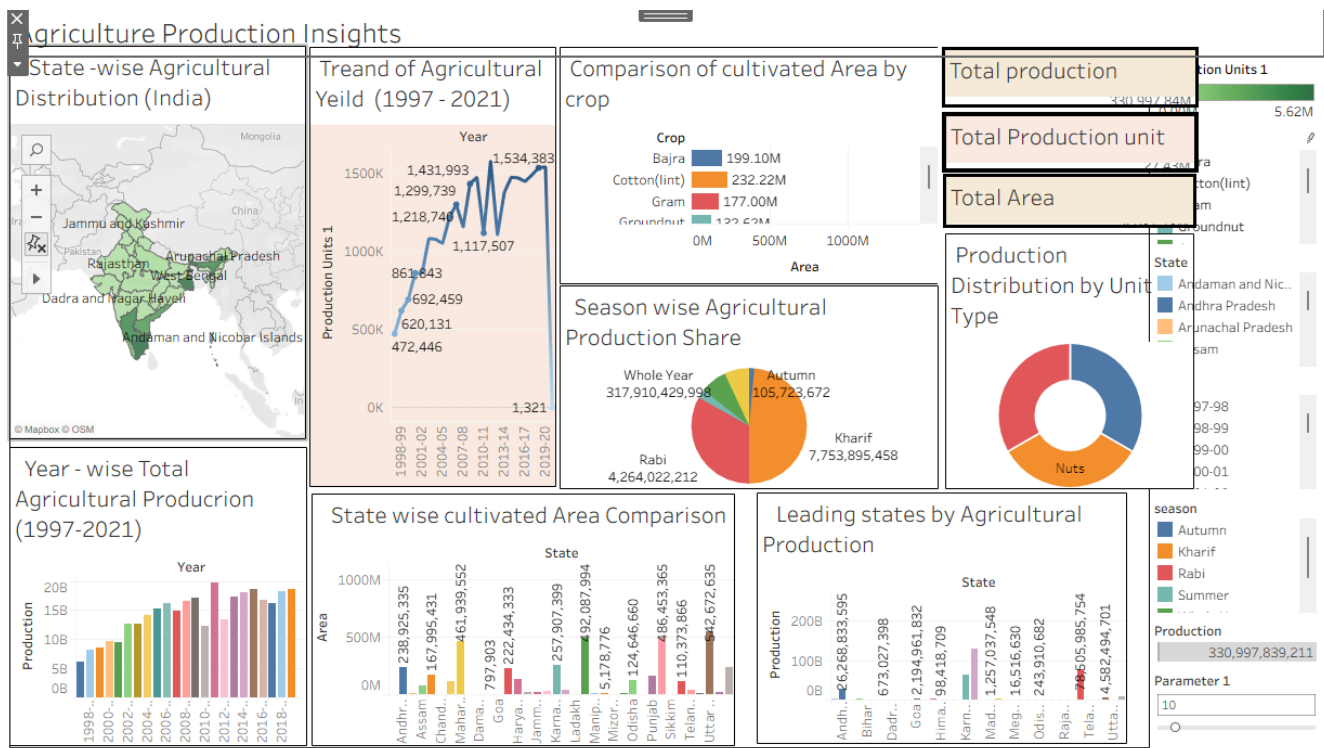


8. How does crop production vary across Indian states?

- **Visualization:** Map showing production by state.
- **Screenshot of visualization:**

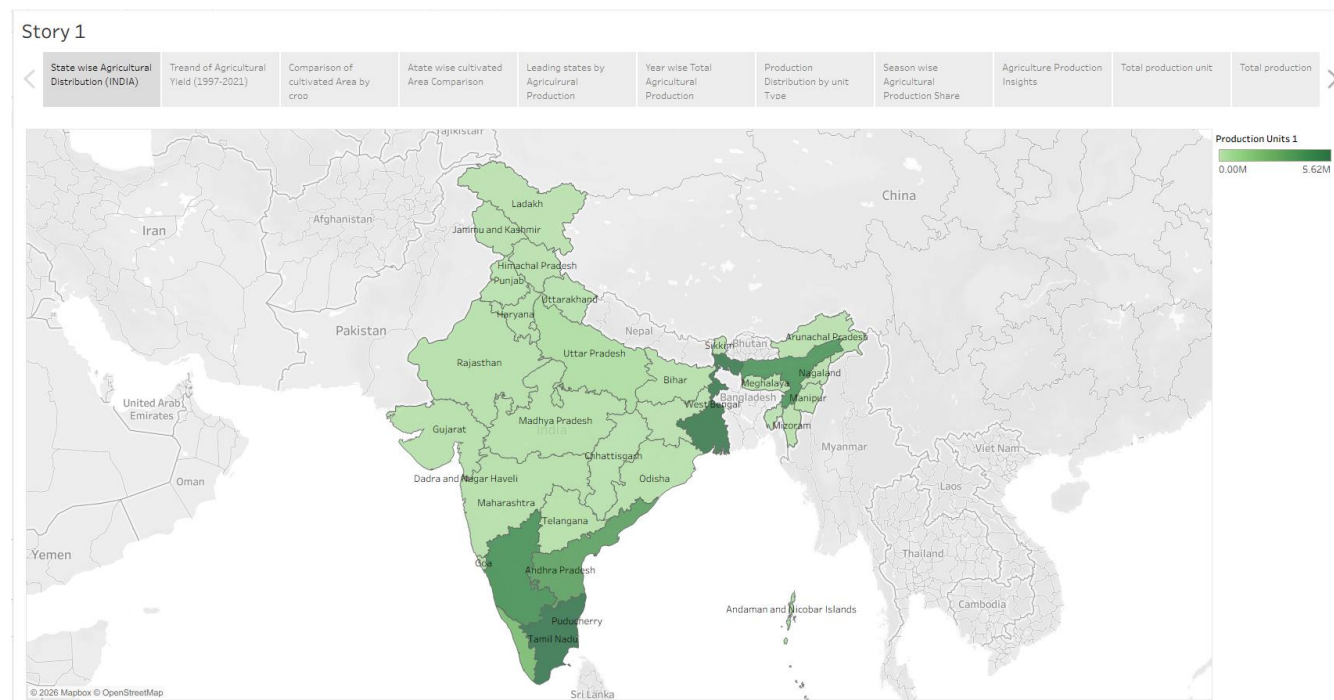
5. Dashboard

5.1. Dashboard Design File



6. Report

6.1. Story Design File



7. Performance Testing

7.1 Utilization of Data filters

Total 5 filters used: 1. State Filter, 2. Year Filter, 3. Crop Filter, 4. Season Filter, 5. Production Range Filter. All filters are working correctly in the dashboard.

7.2 No of Calculation Field

Total 3 Calculated Fields used: 1. Total Production, 2. Total Area Cultivated, 3. Production

Yield. Also used Parameters for dynamic filtering.

7.3 No of Visualization

Total Visualizations: 9 (1 Map + 7 Charts + 1 Donut/Pie). Total Dashboards: 1. Total Stories: 1 with 9 points.

8. Conclusion/Observation

1. Rice is the most cultivated crop in India in terms of area.
2. Karnataka, Tamil Nadu, Madhya Pradesh and West Bengal are leading states in production.
3. Kharif season contributes more than 50% of total production (approx 720M tonnes).
4. Total agricultural production has shown an increasing trend from 1997 to 2019.
5. The data for 2020-21 is incomplete and needs to be corrected.
6. The dashboard provides a one-stop solution for agricultural analysis. e Scope

9. Future Scope

1. Add weather and rainfall data to predict production.
2. Use forecasting to predict production for next 5 years.
3. Add district-level drill-down analysis
4. Connect the dashboard to a live government API for real-time
5. updates. Create a mobile-friendly version of the dashboard.